

Fixed objective weights; adaptive micro-batch counts

(a) Fix w ; allocate m

Fixed task weights

Objective w

fixed 



Math Code Instr. Science

Micro-batches in the next step

Counts m

adaptive



all tasks present

loss $\times w_i/m_i$

$m_i \geq 2$

fixed objective

$$E[A] = \sum_i w_i \nabla L_i$$

(b) Measure update noise

reuse gradients from this step

micro-batch gradient $g_{i,s}$

AdamW coordinates

$$Dg_{i,s}, D = (\sqrt{V} + \epsilon)^{-1}$$

task noise $e_i = \text{Var}(Dg_i)$



no extra forward or backward pass

(c) Allocate compute

reuse the completed step

scaled noise e_i

micro-batch cost τ_i



GPAS

$$m_i \propto w_i \sqrt{e_i}$$

min $V(m)$

optimizer-aware allocation

Cost-GPAS

minimize $J_C(m)$

if $C = 0$: $m_i \propto w_i \sqrt{e_i / \tau_i}$

Evaluate: teacher loss vs. tokens • GPU hours

next step: update m from current noise and cost; keep w fixed